Lecture Scribe: Tail Bounds

L23_Amazone Sale_2026_...
PDF

Generate a lecture scribe for the given PDF file in chat. This is the lecture file that our professor used while conducting the lecture 23. A lecture scribe is an exam-ready reconstruction of what was taught in the class, and it should allow me to recall definitions and notations, reconstruct proofs and derivations, follow worked examples step by step, and understand logical dependencies. Also, a lecture scribe must not have creative explanations, simplified tutorials, textbook rewrites, a solution manual for unseen problems, and a place to add intuition not discussed in class. The key point is that if it was not in the lecture or textbook context, then it should not appear in the lecture scribe. Also, the purpose of lecture scribing is that it must help me during exams to revise concepts in a shorter period. Hence, for the given lecture pdf, make the lecture scribe that also explains from where the particular formula gets derived.

Here is your **exam-ready lecture scribe** reconstructed strictly from the lecture slides:

Lecture 23 Scribe

Predicting the Impossible: Tail Bounds and System Reliability

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1. Problem Context (Amazon Black Friday Case Study)

- System: Large-scale distributed servers (Amazon sale traffic).
- Model: Requests arrive randomly over time → modeled using **Poisson distribution**. (Page 3)

Let:

- X = number of requests in a time interval

- λ = average rate (mean)

From slide:

- Example: $\lambda = 50$ requests/min (Page 5)
- Crash threshold: $k = 60$

Objective

Compute:

$$P(X \geq k)$$

→ Probability system overloads.

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2. Key Observation from Lecture

From *Page 5*:

- "The mean is safe. The tail is catastrophic."

Meaning:

- Average load does not cause failure.
- Rare high spikes (**tail events**) cause crashes.

Thus:

- We must estimate **tail probability** $P(X \geq k)$.

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3. Need for Bounds

Exact probability is difficult or expensive to compute.

From *Page 6*:

- Use **upper bounds** instead of exact values.
- Better information → tighter bounds.

Hierarchy:

1. Markov → uses only mean
2. Chebyshev → uses mean + variance
3. Chernoff → uses full distribution (MGF)

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4. Markov's Inequality

Statement

For non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Derivation (as implied in lecture)

Using indicator:

$$X \geq a \cdot \mathbf{1}_{X \geq a}$$

Take expectation:

$$E[X] \geq a \cdot P(X \geq a)$$

Rearrange:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Properties (Page 7)

- Requires: **Mean only**
- Decay: $\frac{1}{k}$ (linear)
- Very loose (overestimates heavily)

Example from slide:

- Exact ≈ 0.004
- Markov ≈ 0.714 (165× too high)

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5. Chebyshev's Inequality

Statement

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

For right tail:

$$P(X \geq a) \leq \frac{\sigma^2}{(a-\mu)^2}$$

Derivation (based on Markov)

Apply Markov to:

$$Y = (X-\mu)^2$$

Then:

$$P(|X-\mu| \geq t) = P(Y \geq t^2)$$

Using Markov:

$$P(Y \geq t^2) \leq \frac{E[Y]}{t^2}$$

Since:

$$E[(X-\mu)^2] = \sigma^2$$

Final:

$$P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

Properties (Page 8)

- Requires: **Mean + Variance**
- Decay: $\frac{1}{k^2}$ (polynomial)
- Better than Markov

Example:

- Exact ≈ 0.004
- Chebyshev ≈ 0.125 (29× too high)

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6. Chernoff Bound

Key Idea (Page 9)

Use **Moment Generating Function (MGF)**:

$$M_X(t) = E[e^{tX}]$$

General Bound

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Derivation Steps (Page 18 Blueprint)

Step 1: Define exponential transform

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

$$= e^{-ta} M_X(t)$$

Step 3: Optimize over t

$$\min_{t>0} e^{-ta} M_X(t)$$

Poisson Case (Page 9)

For $X \sim Poisson(\lambda)$:

$$M_X(t) = e^{\lambda(e^t - 1)}$$

Final bound form:

$$P(X \geq a) \leq e^{-\lambda(r \ln r - r + 1)}, \quad r = \frac{a}{\lambda}$$

Properties

- Requires: **Full distribution (MGF)**
- Decay: **Exponential** e^{-k}
- Tightest bound

Example:

- Exact ≈ 0.092
- Chernoff ≈ 0.391 (closest)

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7. Comparison of Bounds (Page 10)

Bound	Decay	Tightness
Markov	$1/k$	Very loose
Chebyshev	$1/k^2$	Moderate
Chernoff	e^{-k}	Tight

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8. System Design Interpretation

SLA Scenario (Page 11)

- $\lambda = 500$
- Threshold $k = 600$

Goal:

$$P(X \geq 600)$$

Impact of Bound Choice (Page 12)

Method	Estimated Failure
Exact	~1.4%
Chernoff	~8.8%
Chebyshev	~20%
Markov	~90.9%

Conclusion from Lecture

- Loose bounds → huge overestimation
- Leads to:
  - Over-provisioning
  - Financial waste

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9. Waste Multiplier (Page 14)

$$\text{Waste} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

Interpretation:

- Larger bound ⇒ more unnecessary servers purchased

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10. Distribution Sensitivity (Page 15)

- Chernoff works well for:
  - Poisson
  - Binomial
- Not accurate for:
  - Heavy-tailed (Gamma)

→ In such cases:

- Use simulations or Chebyshev

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11. Decision Matrix (Page 16)

Method	Requires	Use Case
Markov	Mean	Quick rough bound
Chebyshev	Mean + Variance	Unknown distribution
Chernoff	Full MGF	Precise system sizing

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12. Class Exercise (Laplace Distribution)

Given:

$$X \sim Laplace(\mu, b), \quad \sigma = b\sqrt{2}$$

Steps (Page 18)

1. Compute MGF:

$$M_X(t) = E[e^{tX}]$$

2. Apply Chernoff:

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

3. Optimize:

$$\frac{d}{dt}(\cdot) = 0$$

4. Substitute optimal t

Final Takeaways

- Tail probability is critical for system reliability.
- Exact computation often replaced by bounds.
- Tightness hierarchy:

$$\text{Markov} < \text{Chebyshev} < \text{Chernoff}$$

- Wrong bound ⇒ wrong engineering decisions.

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3 Follow-up Questions

1. Why does applying Markov to e^{tX} make Chernoff exponentially tighter?
2. In what scenario would Chebyshev be preferred over Chernoff?
3. How does the Poisson MGF lead to exponential decay in Chernoff bounds?